

Program : Diploma in Communication and Computer Networking	
Course Code : 6321B	Course Title: Digital Communication System
Semester: 6	Credits: 4
Course Category: Engineering Science	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- Understand the elements of Digital Communication System
- Learn the principles of Conversion of Analog signals to Digital Signals
- Learn different digital modulation techniques
- Understand the concepts of multiplexing and multiple access techniques

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge in Electrical and Electronics circuit concepts	2031	Fundamentals of Electrical & Electronics Engineering	1

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Understand The elements of Digital Communication System	14	Understanding
CO2	Learn the principles of Conversion of Analog signals to Digital Signals	14	Applying
CO3	Learn different digital modulation techniques	14	Applying
CO4	Understand the concepts of multiplexing and multiple access techniques	16	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	1				
CO2	3	1	2				
CO3	3	2	1				
CO4	3	2	1				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand The elements of Digital Communication System		
M1.01	To describe role of various elements of digital communication systems	3	Understand
M1.02	To explain various Communication Channel Characteristics	4	Understand
M1.03	To describe Concept of Entropy & Information rate	3	Understand
M1.04	To describe Shannon's Information Capacity Theorem	4	Apply
Contents: Introduction to Digital Communication Systems: Overview of digital communication systems – Advantages of Digital Communication, Communication Channel Types, Characteristics, Modulation Process, Concept of Entropy & Information rate, Shannon's Information Capacity Theorem			
CO2	Learn the principles of Conversion of Analog signals to Digital Signals		
M2.01	To describe the principles of Sampling and Quantization	7	Understanding
M2.02	To describe various analog & Digital Pulse Modulation schemes	7	Understanding
	Series Test –I	1	

Contents:

Digital Coding of Analog Waveforms , Sampling ,Nyquist Sampling Theorem, (only statement) Quantization, Quantization error, companding Analog Pulse Modulation, . PAM, PPM, PWM Digital Pulse Modulation Pulse-Code Modulation, Differential Pulse-Code Modulation ,Delta Modulation

CO3	Learn different digital modulation techniques		
M3.01	To describe Concept of coherent & Non coherent detection	3	Understanding
M3.02	To describe principles of Generation and Detection of BASK, BFSK, BPSK &QPSK	5	Understanding
M3.03	To describe the working of BASK, BFSK, BPSK &QPSK with waveforms..	6	Applying

Contents:

Digital modulation schemes: Concept of coherent & Non coherent detection ,Binary ASK, FSK, PSK, and QPSK General Principles of Production and Detection with waveforms.

CO4	Understand the concepts of multiplexing and multiple access techniques		
M 4.01	To describe different types of multiplexing	8	Understanding
M 4.02	To describe different types of multiple access techniques	8	Understanding
	Series Test – II	1	

Contents:

Need and method of multiplexing: Time Division Multiplexing(TDM), Frequency division Multiplexing(FDM) Code division Multiplexing, Need and method of access techniques :Time Division Multiple access (TDMA), Frequency division Multiple access (FDMA) Code division Multiple access (CDMA) – Introduction to Spread Spectrum modulation.

Text / Reference

T/R	Book Title/Author
T1	Electronic Communication Systems- Tomasi, Wayne
T2	Digital Communication – P Ramakrishna Rao
T3	Communication Systems – Simon Haykin
T4	Digital Communication – Sklar & Bernald

Online Resources

Sl.No	Website Link
1	Video lecture:- www.nptelvideos.in/communication engineering .
2	Digital Modulation technique:- https://www.youtube.com/watch?v=GLnGVB92K78
3	Multiple access:- https://www.youtube.com/watch?v=vtiuplw1c4E
4	CDMA:- https://www.youtube.com/watch?v=vdbc9P3U-Xo